Program 7: Deploy a Web Application to Kubernetes using Minikube

Project Structure

program7
|--nginx-deployment.yaml

Step 1: Start the minikube

```
isha@meow: ~/DevOps/program7
isha@meow:~/DevOps/program7$ minikube start
😄  minikube v1.37.0 on Ubuntu 24.04
✨  Using the docker driver based on existing profile
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.48 ...
🔄  Restarting existing docker container for "minikube" ...
🐳  Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔎  Verifying Kubernetes components...
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
    ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
    ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
💡  Some dashboard features require the metrics-server addon. To enable all features please run:

        minikube addons enable metrics-server

🌟  Enabled addons: storage-provisioner, default-storageclass, dashboard
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
isha@meow:~/DevOps/program7$
```

Step 2: Check the status of minikube

```
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isha@meow: ~/DevO
isha@meow:~/DevOps/program7$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Step 3: Create deployment file
nano nginx-deployment.yaml

Step 4: Apply the nginx-deployment.yaml

```
isha@meow:~/DevOps/program7$ cat > nginx-deployment.yaml <<'EOF'
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
EOF
isha@meow:~/DevOps/program7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
isha@meow:~/DevOps/program7$ kubectl get deployments
NAME               READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment   0/2     2            0           11s
```

Step 5: Check status
Step 6: Expose the deployment as a Service

```
isha@meow:~/DevOps/program7$ kubectl get pods -l app=nginx
NAME                                READY   STATUS              RESTARTS   AGE
nginx-deployment-6f9664446b-n5tkx   0/1     ContainerCreating   0          18s
nginx-deployment-6f9664446b-nzg2z   0/1     ContainerCreating   0          18s
isha@meow:~/DevOps/program7$ kubectl expose deployment nginx-deployment --type=NodePort --port=80
service/nginx-deployment exposed
isha@meow:~/DevOps/program7$ kubectl get services
NAME               TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
kubernetes         ClusterIP   10.96.0.1       <none>        443/TCP        3m18s
nginx-deployment   NodePort    10.110.60.134   <none>        80:30447/TCP   37s
isha@meow:~/DevOps/program7$ minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:30447

Step 7: Check service

```
isha@meow:~/DevOps/program7$ kubectl get svc
NAME            TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
kubernetes      ClusterIP   10.96.0.1        <none>        443/TCP        14d
nginx-service   NodePort    10.106.100.149   <none>        80:32681/TCP   7d
isha@meow:~/DevOps/program7$
```

Step 8: Access the Web Application